Multiplicative-gap approximate counting by shallow logspace-uniform AC circuits

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Allowed standard prerequisites (editorial).

- Ajtai constant-depth approximate majority. Author Lemma15 explicitly cites [2], full quoted statement retained.
- Convolution-based pairwise independent hashing. Author Definition12/Theorem13, construction attributed to [22]; exact construction and guarantee retained.

Author circuit and hashing definitions

Definition 8 (AC circuit). AC circuit is a circuit with input gates, NOT gates, unbounded fan-in AND/OR gates, and (possibly more than one) output gates. The size of a circuit is defined by the number of AND/OR gates. The depth of a circuit is defined by the largest number of AND/OR gates on any path from an input gate to an output gate.

Definition 9 (L-uniform AC circuit family). For functions $S, d : \mathbb{Z}^+ \to \mathbb{R}^+$, we say a collection of circuits $\{C_n\}_{n\in\mathbb{Z}^+}$ is an S-size d-depth L-uniform AC circuit family, if each C_n has size $\leq S(n)$ and depth $\leq d(n)$, and given the binary representation of n, the description of C_n can be computed in uniform $O(\log n)$-space. We need to mention that the number of input gates in C_n is not necessarily n. Also note that since we can encode a tuple of $O(1)$ many integers to a single integer, we can also consider circuit collections with a tuple of integers as an index.

Definition 10 (Complexity Class L-AC^1). We say a language L is in the class L-AC^1 if there exists a poly(n)-size $O(\log n)$-depth L-uniform AC circuit family $\{C_n\}$ such that C_n computes L on n-bit inputs.

Definition 11 (GapMaj). For $n \in \mathbb{Z}^+$ and $a, b \in \mathbb{R}$ such that $0 \leq a < b \leq n$, define the promise problem GapMaj$[a, b]$ on n bits as follow:

$$\text{GapMaj}[a,b](x_1,\ldots,x_n) := \begin{cases} \text{YES} & \text{if } x_1,\ldots,x_n \text{ contains } \geq b \text{ 1's} \\ \text{NO} & \text{if } x_1,\ldots,x_n \text{ contains } \leq a \text{ 1's} \\ \perp & \text{otherwise.} \end{cases}$$

We will use pairwise independent hash function as a tool for approximate counting in AC. We shall use the following construction based on convolution, which was also used in [22].

Definition 12 (Convolution-Based Pairwise Independent Hash Function). Suppose m is a power of 2. Define $H_m : [m^3] \times [m] \to [m]$ by: for $(k, x) \in [m^3] \times [m]$, let $x_1 \cdots x_{\log m}$ be the binary representation of $x - 1$, let $a_1 \cdots a_{2\log m} b_1 \cdots b_{\log m}$ be the binary representation of $k - 1$, let $y_j := (\sum_{i=1}^{\log m} a_{i+j} x_i + b_j) \bmod 2$ for $j \in [\log m]$, then define $H_m(k, x)$ by letting $y_1 \cdots y_{\log m}$ be the binary representation of $H_m(k, x) - 1$.

Theorem 13. H_m is Pairwise Independent Hash Function in the following sense: for any $1 \le i < j \le m$, when k is sampled from the uniform distribution over $[m^3]$, the joint distribution of $(H_m(k, i), H_m(k, j))$ is identical to the uniform distribution over $[m] \times [m]$.

Theorem 14. Let $n, a, b \in \mathbb{N}$ such that $0 \le a < b \le n$. Then there exists a poly(n)-size $O\left(\frac{\log \frac{b}{b-a}}{\log \log n} + 1\right)$-depth L-uniform AC circuit family $\{C_{n,a,b}\}$ that computes GapMaj$[a, b]$ on n bits.

The proof depends on the next few Lemmas.

Lemma 15 ([2]). Let $n \in \mathbb{Z}^+$. Then there exists poly(n)-size $O(1)$-depth L-uniform AC circuit family $\{C_n^{(0)}\}$ that computes GapMaj$[n/3, 2n/3]$ on n bits.

Lemma 16 (Exact Counting). Let $n, \ell \in \mathbb{Z}^+$ such that $n \ge \ell$. Then there exists a poly(n)-size $O\left(\frac{\log \ell}{\log \log n} + 1\right)$-depth L-uniform AC circuit family $\{E_{n,\ell}\}$ such that on ℓ bits of input, $E_{n,\ell}$ outputs the exact number of 1's over the input bits, in binary form. We remark that in Lemma 16, n is only used to bound the size of the circuit. Also, n, ℓ are not necessarily polynomially related.

Proof. We only need to show how to compute sum of $O(\sqrt{\log n})$ many $O(\log n)$-bit[1] non-negative integers in $O(1)$-depth, then by divide-and-conquer we can compute sum of ℓ bits in $O\left(\frac{\log \ell}{\log \log n} + 1\right)$-depth. View the $O(\log n)$-bit integers as $2^{\lfloor \sqrt{\log n} \rfloor}$-base $O(\sqrt{\log n})$-digit integers. We use the grade-school algorithm to sum $O(\sqrt{\log n})$ integers as follow. We first guess the result and all carry-bits, which involve at most $O(\sqrt{\log n}) \cdot O\left(\log\left(\sqrt{\log n} \cdot 2^{\lfloor \sqrt{\log n} \rfloor}\right)\right) = O(\log n)$ bits, and thus has at most poly(n) choices. Then we can apply a local check on each digit, each local check involves at most $O(\log n)$ bits, and thus deciding whether all local checks are passed can be computed in $O(1)$-depth. Then we can take the result of the only guess that passes all local checks. The total cost is $O(1)$-depth. □

[1] Here "$O(\log n)$-bit integers" refers to integers which has $O(\log n)$-bits in its binary representation.

Lemma 17. Let $n, a, b \in \mathbb{N}$ such that $0 \le a < b \le n$. Then there exists a poly(n)-size $O\left(\frac{\log \frac{n}{b-a}}{\log\log n} + 1\right)$-depth L-uniform AC circuit family $\{C^{(1)}_{n,a,b}\}$ that computes GapMaj$[a, b]$ on n bits.

Proof. Only consider the case that n is a power of 2, otherwise we can use a simple padding argument. By Lemma 15, it suffices to show how to reduce GapMaj$[a, b]$ on n bits to GapMaj$[n^3/3, 2n^3/3]$ on n^3 bits, via a poly(n)-size $O\left(\frac{\log \frac{n}{b-a}}{\log\log n} + 1\right)$-depth L-uniform AC circuit. If $b - a \le 4\sqrt{n}$ then we can directly compute the number of 1's exactly via Lemma 16. Below we only consider $b - a > 4\sqrt{n}$. Let $\ell := \left\lceil \frac{12n^2}{(b-a)^2} \right\rceil$. Suppose the GapMaj$[a, b]$ instance is $x_1, x_2, \ldots, x_n$. Let H_n be the hash function defined in Definition 12. Define $y_1, \ldots, y_{n^3}$ as follow: for $i \in [n^3]$, let y_i be 1 if at least $\frac{a+b}{2n}$ fraction of $x_{H_n(i,1)}, \ldots, x_{H_n(i,\ell)}$ is 1, otherwise let y_i be 0. Note that $y_1, \ldots, y_{n^3}$ can be computed via a poly(n)-size $O\left(\frac{\log \ell}{\log\log n} + 1\right)$-depth L-uniform AC circuit, by Lemma 16. Here $O\left(\frac{\log \ell}{\log\log n} + 1\right) = O\left(\frac{\log \frac{n}{b-a}}{\log\log n} + 1\right)$. Let's do some simple calculations. Assume p fraction of $x_1, \ldots, x_n$ is 1. Let S_i be number of 1's in $x_{H_n(i,1)}, \ldots, x_{H_n(i,\ell)}$. Then we have $\mathbb{E}_{i\sim[n^3]}[S_i] = p\ell$ and $\mathrm{Var}_{i\sim[n^3]}[S_i] \le \ell$. So if $p \le \frac{a}{n}$, then

$$\Pr_{i\sim[n^3]}\left[S_i \ge \ell \cdot \frac{a+b}{2n}\right] \le \frac{\ell}{(\ell \cdot \frac{b-a}{2n})^2} = \frac{4n^2}{\ell(b-a)^2} \le \frac{1}{3}.$$

Similarly if $p \ge \frac{b}{n}$ then

$$\Pr_{i\sim[n^3]}\left[S_i \le \ell \cdot \frac{a+b}{2n}\right] \le \frac{1}{3}.$$

This means if $x_1, \ldots, x_n$ is YES/NO instance of GapMaj$[a, b]$, then $y_1, \ldots, y_{n^3}$ is YES/NO instance of GapMaj$[n^3/3, 2n^3/3]$. The reduction is completed. □

Proof of Theorem 14. We will try to reduce to Lemma 17. Suppose the GapMaj$[a, b]$ instance is $x_1, x_2, \ldots, x_n$. We only consider the case n is a power of 2, otherwise use a simple padding argument. We only consider the case $10\left(\frac{b}{b-a}\right)^2 < \frac{n}{b-a}$ (or equivalently, $n(b-a) > 10b^2$), otherwise we can directly apply Lemma 17. Let $\ell := \left\lfloor \frac{n(b-a)}{2b^2} \right\rfloor$. For $i \in [n^3]$, let $y_i := x_{H_n(i,1)} \vee \cdots \vee x_{H_n(i,\ell)}$, here H_n is the hash function defined in Definition 12. Then $y_1, \ldots, y_{n^3}$ can be computed via poly(n)-size $O(1)$-depth L-uniform AC circuit. Assume p fraction of $x_1, \ldots, x_n$ is 1. Let S_i be number of 1's in $x_{H_n(i,1)}, \ldots, x_{H_n(i,\ell)}$. Then we have $\mathbb{E}_{i\sim[n^3]}[S_i] = p\ell$ and $\mathbb{E}_{i\sim[n^3]}[S_i^2] = \ell(\ell-1)p^2 + \ell p \le \ell p + \ell^2 p^2$. Thus by

$$\frac{\mathbb{E}_{i\sim[n^3]}[S_i]^2}{\mathbb{E}_{i\sim[n^3]}[S_i^2]} \le \Pr_{i\sim[n^3]}[S_i \ge 1] \le \mathbb{E}_{i\sim[n^3]}[S_i]$$

we know: if $p \le \frac{a}{n}$, then $\Pr_{i\sim[n^3]}[S_i \ge 1] \le \frac{\ell a}{n}$; if $p \ge \frac{b}{n}$, then

$$\Pr_{i\sim[n^3]}[S_i \ge 1] \ge \frac{(\frac{\ell b}{n})^2}{\frac{\ell b}{n} + (\frac{\ell b}{n})^2} \ge \frac{\ell b}{n} - \left(\frac{\ell b}{n}\right)^2.$$

To summarize, if $x_1, \ldots, x_n$ is YES/NO instance of GapMaj$[a, b]$, then $y_1, \ldots, y_{n^3}$ is YES/NO instance of

$$\text{GapMaj}\left[\left\lfloor n^3 \cdot \frac{\ell a}{n}\right\rfloor, \left\lceil n^3 \cdot \left(\frac{\ell b}{n} - \left(\frac{\ell b}{n}\right)^2\right)\right\rceil\right].$$

Finally we observe that

$$\left(\frac{\ell b}{n} - \left(\frac{\ell b}{n}\right)^2\right) - \frac{\ell a}{n} = \ell \cdot \left(\frac{b-a}{n} - \frac{\ell b^2}{n^2}\right) \geq \frac{n(b-a)}{3b^2} \cdot \frac{b-a}{2n} = \frac{(b-a)^2}{6b^2}.$$

Thus by Lemma 17, GapMaj $\left[\left\lfloor n^3 \cdot \frac{\ell a}{n}\right\rfloor, \left\lceil n^3 \cdot \left(\frac{\ell b}{n} - \left(\frac{\ell b}{n}\right)^2\right)\right\rceil\right]$ over n^3 bits can be computed via a poly(n)-size $O\left(\frac{\log \frac{b}{b-a}}{\log\log n} + 1\right)$-depth L-uniform AC circuit. □

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